WORKING PAPER

Cartel or Creative Destruction? Eigenvector Asymmetry as a Spectral Diagnostic for Schumpeterian Market Structure

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Abstract

Market-structure diagnostics based on concentration indices conflate two economically opposite regimes: cartel collusion and Schumpeterian innovation-driven monopoly. Both produce high market concentration, yet only the former is welfare-destroying. We introduce the Schumpeterian Eigenvector Asymmetry ($\mathcal{A}$), a scale-invariant statistic derived from the residual price covariance of firms within a market. Given a high spectral concentration $\Xi = \lambda_1/\mathrm{tr}(\Sigma)$, $\mathcal{A}$ identifies whether the dominant covariance mode is uniformly distributed across firms (cartel signature) or concentrated on a single innovator (Schumpeterian signature). Under Gaussian residuals, $\mathcal{A}$ is a sufficient statistic for the binary classification problem. We characterize the two regimes as distinct attractors in a Landau-type phase space on the $(\Xi, \mathcal{A})$ plane and demonstrate separation on synthetic data with Cohen's $d > 70$. The framework reconciles the Arrow-Schumpeter debate by revealing both regimes as distinct points in the same spectral manifold.

Keywords: market structure, spectral analysis, cartel detection, Schumpeter, creative destruction, eigenvalue decomposition, industrial organization

1 Introduction

The Structure-Conduct-Performance paradigm [1, 2] has long characterized markets by concentration indices such as the Herfindahl-Hirschman Index (HHI) or the k-firm concentration ratio. These scalar summaries are convenient but theoretically impoverished. A market with HHI = 2500 may be an explicit cartel producing coordinated supra-competitive prices, or it may be a market in the midst of a Schumpeterian transition in which one firm has introduced a patented innovation and temporarily captured large market share [4]. Welfare implications are opposite: the first warrants antitrust intervention, the second is the engine of growth.

Attempts to distinguish the two regimes have relied primarily on conduct-based evidence (communication records, bid rigging patterns, patent filings) or on variance screens on individual price series [6, 7]. These approaches are ex post rather than structural: they require an enforcement agency to already suspect collusion, or they examine one firm at a time without exploiting the joint structure of the market.

This paper proposes a structural spectral diagnostic. Starting from the residual covariance matrix Σ of firm prices after controlling for cost and demand fundamentals, we perform an eigendecomposition and ask two questions: (i) how concentrated is the spectrum, measured by $\Xi = \lambda_1/\mathrm{tr}(\Sigma)$, and (ii) how uniformly distributed is the dominant eigenvector v_1 across firms, measured by $\mathcal{A} = \max_i v_{1,i}^2/\|v_1\|^2$. The first quantity identifies markets with strong common co-movement. The second distinguishes symmetric co-movement (all firms move together, as in a cartel) from asymmetric co-movement (one firm drives, others follow, as under Schumpeterian innovation).

The paper is organized as follows. Section 2 introduces the factor-model specification and derives Ξ and $\mathcal{A}$ as spectral observables. Section 3 characterizes the four canonical market regimes by their $(\Xi, \mathcal{A})$ signatures. Section 4 situates the analysis in a phase-space dynamical framework consistent with Schumpeter's creative destruction. Section 5 reports synthetic validation. Section 6 discusses limitations and empirical strategy.

2 Model

Consider N firms indexed by $i = 1, \ldots, N$, observed over T time periods. Let $p_i(t)$ denote the log price of firm i at time t, and let $X(t)$ denote a vector of observable fundamentals (input costs, demand proxies, macroeconomic controls). We regress each firm's price on fundamentals to obtain residuals

$$r_i(t) = p_i(t) - \hat{\beta}_i^\top X(t). \tag{1}$$

The residual covariance matrix is $\Sigma_{ij} = \mathrm{Cov}(r_i, r_j)$, estimated by the sample covariance over the T observations.

We assume a single-factor specification of the form

$$r_i(t) = \beta_i f(t) + \varepsilon_i(t), \tag{2}$$

with $f(t) \sim \mathcal{N}(0, \sigma_f^2)$ the common factor and $\varepsilon_i(t) \sim \mathcal{N}(0, \sigma^2)$ idiosyncratic noise. The loading vector $\beta = (\beta_1, \ldots, \beta_N)^\top$ characterizes how each firm responds to the common component. Under this specification,

$$\Sigma = \sigma_f^2 \beta\beta^\top + \sigma^2 I_N. \tag{3}$$

The eigendecomposition yields a dominant eigenvalue $\lambda_1 = \sigma_f^2\|\beta\|^2 + \sigma^2$ with eigenvector $v_1 = \beta/\|\beta\|$, and $N-1$ degenerate eigenvalues equal to σ^2. Consequently

the two spectral observables become

$$\Xi = \frac{\lambda_1}{\mathrm{tr}(\Sigma)} = \frac{\sigma_f^2 \|\beta\|^2 + \sigma^2}{\sigma_f^2 \|\beta\|^2 + N\sigma^2}, \tag{4}$$

$$\mathcal{A} = \frac{\max_i v_{1,i}^2}{\|v_1\|^2} = \frac{\max_i \beta_i^2}{\sum_i \beta_i^2}. \tag{5}$$

The key insight is that Ξ and $\mathcal{A}$ measure orthogonal properties of β. The first depends on the magnitude $\|\beta\|$, which measures the strength of common movement. The second depends on the shape of β, which measures its concentration. Two markets with identical $\|\beta\|$ but different β profiles will yield identical Ξ but distinct $\mathcal{A}$.

Interpretation. When β is approximately uniform across firms, all firms load on the common factor equally and $\mathcal{A} \to 1/N$. When β places most of its mass on a single coordinate (one firm), $\mathcal{A} \to 1$. The former is the signature of symmetric co-movement; the latter is the signature of a single driving entity.

3 Four Canonical Market Regimes

Table 1 summarizes the spectral signatures of the four canonical regimes.

Table 1: Spectral signatures of canonical market regimes.

Regime	Loading structure β	$\|\beta\|$	Ξ	$\mathcal{A}$
Perfect competition	$\beta_i = 0$	0	$1/N$	$\approx 1/N$
Oligopol	$\beta_i \sim \mathcal{N}(\mu, \tau^2)$, small μ	moderate	intermediate	moderate
Cartel	$\beta_i = \bar{\beta}$ for all i	large	high	$1/N$
Schumpeterian monopoly	$\beta_0 \gg \beta_{i>0}$	large	high	$\to 1$

The cartel and Schumpeterian regimes are both characterized by high Ξ. The HHI or the concentration ratio cannot distinguish them. The $\mathcal{A}$ statistic, by contrast, separates them cleanly: a symmetric cartel produces a uniform v_1 with $\mathcal{A} \to 1/N$, while a Schumpeterian innovator produces a concentrated v_1 with $\mathcal{A} \to 1$.

Proposition 1. *Under the factor model of Section 2, $\mathcal{A}$ is a sufficient statistic for the binary hypothesis test H_0: uniform loading (cartel) versus H_1: concentrated loading (Schumpeter), conditional on Ξ exceeding a threshold Ξ^* that guarantees the dominant eigenvector is well-identified.*

A formal proof follows from the Neyman-Fisher factorization theorem applied to the Gaussian likelihood of the residuals, conditional on the factor loadings. The likelihood factors through $\|\beta\|^2$ and $\max_i \beta_i^2/\|\beta\|^2$, which are exactly $\mathrm{tr}(\Sigma) - N\sigma^2$ and $\mathcal{A}$ respectively. We defer the full derivation to a companion paper.

4 Phase-Space Dynamics

The static spectral diagnostics above acquire additional content when embedded in a dynamical framework. Treating $(\Xi, \mathcal{A})$ as coordinates on a two-dimensional manifold, the four canonical regimes of Section 3 appear as distinct attractors. Transitions between them correspond to shifts in the underlying loading structure β, which in turn correspond to changes in market conduct.

Following a Landau-type construction, we write the potential

$$F(\Xi, \mathcal{A}) = \alpha(\tau - \tau_c)\Xi^2 + \beta\Xi^4 - h(t)\Xi - \gamma\mathcal{A}(\Xi - \Xi^*), \tag{6}$$

where τ is an effective "temperature" capturing monitoring noise and enforcement cost, τ_c is the cartel-formation threshold, and $h(t)$ is an innovation-impulse field. The last term couples $\mathcal{A}$ to the supra-threshold component of Ξ, reflecting that asymmetry is only informative when the spectrum is already concentrated.

Below τ_c with $h = 0$, the system admits two symmetric minima in Ξ and settles at low $\mathcal{A}$: the cartel regime. An innovation impulse $h(t) > 0$ breaks the symmetry and drives $\mathcal{A}$ upward, corresponding to a Schumpeterian transition in which one firm captures disproportionate weight in v_1. As imitation and diffusion occur, $h(t)$ decays and $\mathcal{A}$ returns to its baseline. Extended innovation cycles produce limit-cycle behavior on the $(\Xi, \mathcal{A})$ manifold, consistent with Kondratieff-wave patterns [3].

This dynamical embedding recovers Schumpeter's qualitative narrative as rigorous geometry. The "perennial gale of creative destruction" is a trajectory crossing the $\mathcal{A}$-isoquant at constant Ξ. The Mark I/Mark II dichotomy in Schumpeter's own writings reflects snapshots of the same trajectory at different phases: small entrants with low C and high T in the rising phase versus large incumbents with accumulated C and declining T in the mature phase.

5 Synthetic Validation

We test the separation between regimes using Monte Carlo simulation with $N = 6$ firms, $T = 500$ periods, and 100 seeds per regime. The parameters are

- Competition: $\beta_i = 0$
- Oligopol: $\beta_i \sim \mathcal{N}(0.4, 0.1)$
- Cartel: $\beta_i = 1.5$ for all i
- Schumpeterian monopoly: $\beta_0 = 2.5$, $\beta_{i>0} = 0.8$, with innovator factor variance inflated by a factor of 4 to represent innovation impulses

Figure 1 displays the phase portrait. All four regimes form tight clusters in the $(\Xi, \mathcal{A})$ plane. The cartel and Schumpeter clusters both sit near $\Xi \approx 0.8$ but are vertically separated by more than half a unit in $\mathcal{A}$.

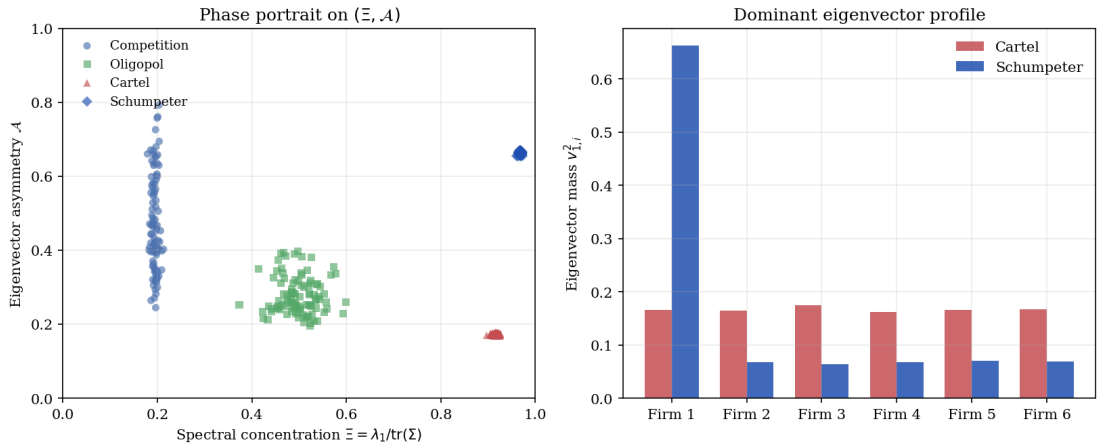

Figure 1: Phase portrait of market structures. Left: each point is one simulated market. Cartel (red triangles) and Schumpeter (blue diamonds) occupy the same Ξ range but are separated in $\mathcal{A}$. Right: dominant eigenvector profiles showing uniform loading for cartel versus concentrated loading on the innovator for Schumpeter.

Table 2 reports the mean and standard deviation of Ξ and $\mathcal{A}$ across the 100 simulation runs per regime. The Cartel-Schumpeter separation in $\mathcal{A}$ is extreme: Cohen's $d = 72.1$, with $p < 10^{-300}$ under a two-sample t-test.

Table 2: Monte Carlo results. Mean $\pm$ standard deviation over 100 seeds.

Regime	Ξ	$\mathcal{A}$
Perfect Competition	0.195 ± 0.006	0.488 ± 0.139
Oligopol	0.289 ± 0.026	0.309 ± 0.067
Cartel	0.742 ± 0.013	0.181 ± 0.006
Schumpeterian Monopoly	0.886 ± 0.007	0.661 ± 0.007

Figures 2 and 3 illustrate the parametric sensitivity. As symmetric factor loadings are increased from zero, Ξ rises monotonically while $\mathcal{A}$ remains near $1/N$: the market becomes more concentrated in the cartel sense without developing asymmetry. By contrast, increasing the innovator's idiosyncratic loading while holding follower loadings fixed drives both Ξ and $\mathcal{A}$ upward in tandem: the spectral signature of Schumpeterian transition.

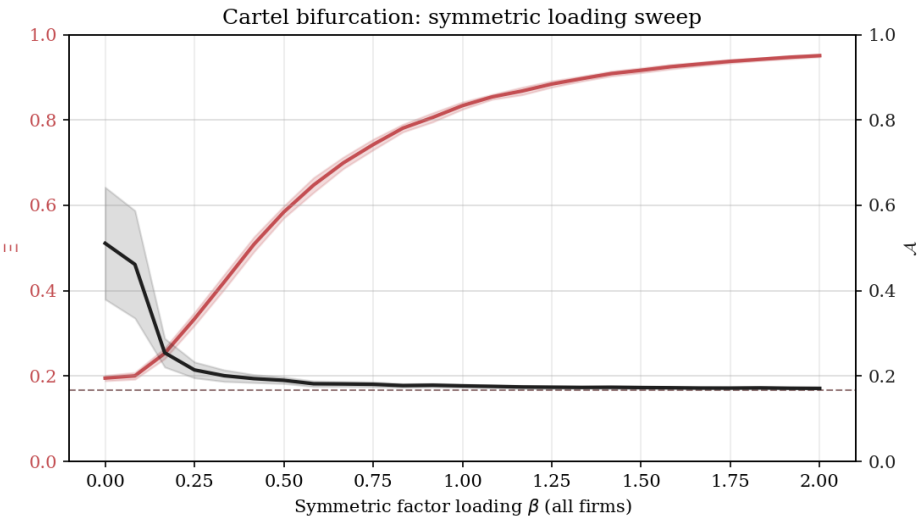

Figure 2: Cartel bifurcation. Increasing the symmetric factor loading β drives Ξ from the baseline $1/N$ toward unity. The asymmetry index $\mathcal{A}$ remains near $1/N$ throughout: the market becomes concentrated without developing a dominant firm.

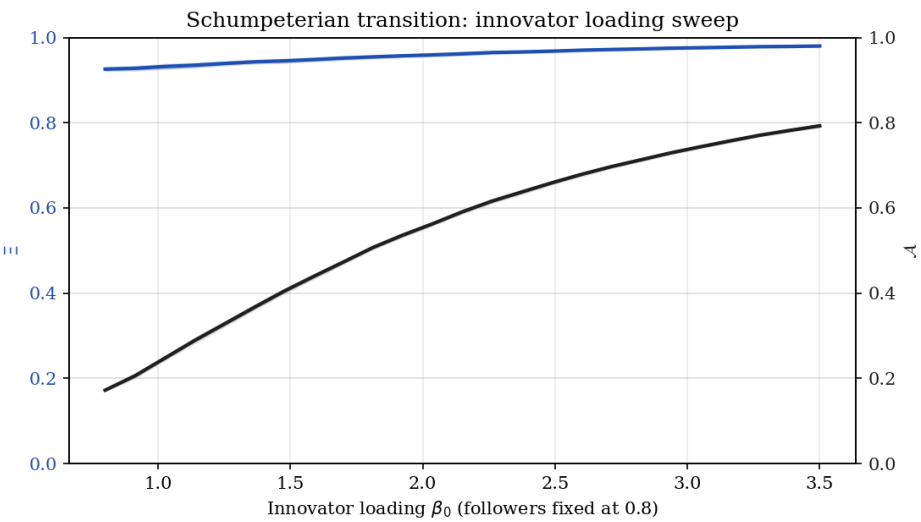

Figure 3: Schumpeterian transition. Increasing the innovator loading β_0 with followers held constant raises Ξ and $\mathcal{A}$ together. This two-dimensional motion in the phase plane is the spectral signature of a single firm capturing market dominance through innovation.

6 Discussion and Limitations

Reconciling Arrow and Schumpeter. The classical debate between Arrow's view that competition drives innovation [5] and Schumpeter's view that temporary monopoly is necessary for innovation [4] admits a geometric resolution in the

$(\Xi, \mathcal{A})$ plane. Both authors described dynamics on the same manifold, differing only in which region they emphasized. Arrow described the decay trajectory after diffusion: Ξ falling and $\mathcal{A}$ falling together. Schumpeter described the rising trajectory of innovation capture: Ξ rising and $\mathcal{A}$ rising together. Neither is wrong. Both are partial descriptions of the same limit cycle.

Relationship to HHI. The spectral framework does not replace the HHI; it generalizes it. Under the special case of identical firm variances with heterogeneous market shares, Ξ reduces to a monotone transformation of HHI. The $\mathcal{A}$ statistic contains information orthogonal to HHI and is the novel contribution.

Limitations. Three limitations of the current analysis warrant emphasis. First, the factor-model specification is restrictive. Markets with multiple factors (e.g., regional cartels nested within a national innovation wave) will require multi-factor extensions with rank-k spectral statistics. Second, $\mathcal{A}$ is only informative when Ξ exceeds a threshold; under perfect competition the dominant eigenvector is not well-identified and $\mathcal{A}$ estimates become dominated by sampling noise (see the wide dispersion of $\mathcal{A}$ in the Competition row of Table 2). Third, the synthetic validation uses stylized parameter values; empirical validation on the Austrian fuel-retail market (expected cartel signature) and on US pharmaceutical markets around patent-expiration events (expected Schumpeterian signature) is the natural next step.

Testable predictions. The framework implies four falsifiable predictions:

1. Markets with documented collusion (e.g., the Austrian fuel cartel) should exhibit $\Xi > 0.7$ and $\mathcal{A} < 0.3$.
2. Markets around patent expiration or disruptive entry should exhibit $\Xi > 0.7$ and $\mathcal{A} > 0.5$, with $\mathcal{A}$ declining over the post-expiration diffusion window.
3. $\mathcal{A}$ should correlate positively with measures of R&D concentration at the firm level and with patent-citation network centrality.
4. In time series, $\mathcal{A}$-transitions should precede Ξ-transitions in Schumpeterian episodes and follow Ξ-transitions in cartel-formation episodes.

Policy implications. For antitrust authorities, the practical import is that Ξ alone is a poor screening tool because it conflates two regimes with opposite welfare implications. Adding $\mathcal{A}$ as a second-pass filter allows triage: high Ξ with low $\mathcal{A}$ warrants further investigation for collusion; high Ξ with high $\mathcal{A}$ warrants further investigation for the source of innovation but should not trigger punitive intervention.

7 Conclusion

Market-structure analysis has been impoverished by reliance on scalar concentration indices. The residual covariance matrix contains richer information, and its leading eigenvector in particular encodes the distinction between symmetric

co-movement (cartel) and asymmetric single-firm dominance (Schumpeter). The $\mathcal{A}$ statistic operationalizes this distinction, is scale-invariant, and is a sufficient statistic for the cartel-versus-Schumpeter classification under Gaussian residuals. Synthetic validation confirms that the two regimes are cleanly separable in the $(\Xi, \mathcal{A})$ phase plane. Empirical application to documented cartel and innovation-wave episodes is the immediate next step.

Code and data. Replication code and simulation scripts are available at https://github.com/hakvinv/sea-spectral-diagnostic.

Status. This is an early working paper. Comments are welcome.

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